Machine Learning for teens

By Hamid Sadeghi

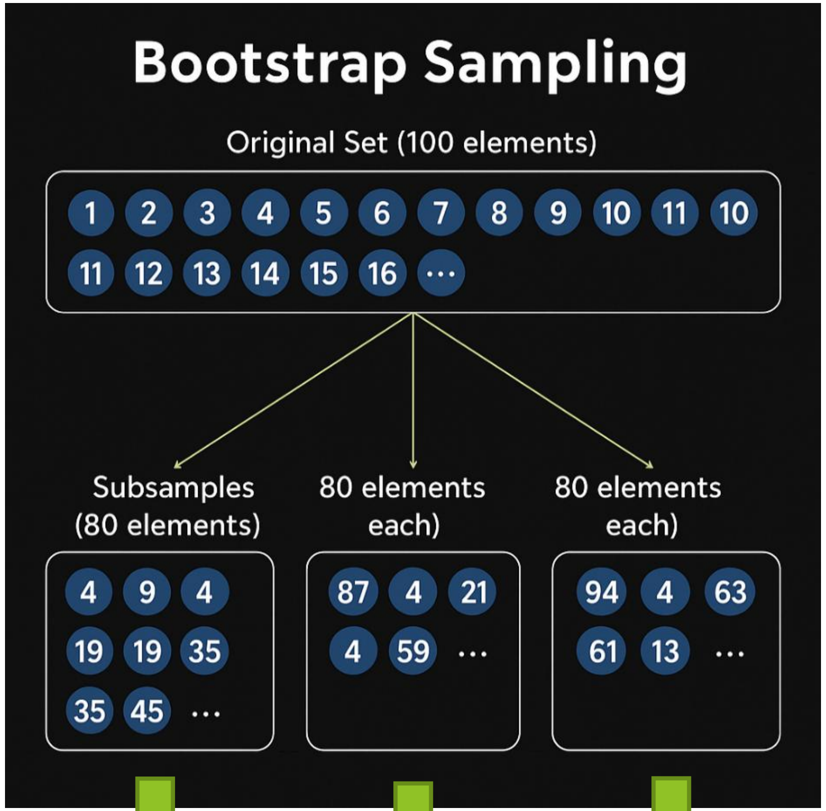
Table of contents

Comparison of Ensemble Learning Methods(review)

Feature	Bagging	Boosting	Stacking
Training Style	Parallel	Sequential	Parallel
Base Models	Same type	Same type	Different types
Data Used	Bootstrapped subsets	Full data	Full data

Bagging

Bootstrap



Aggregation

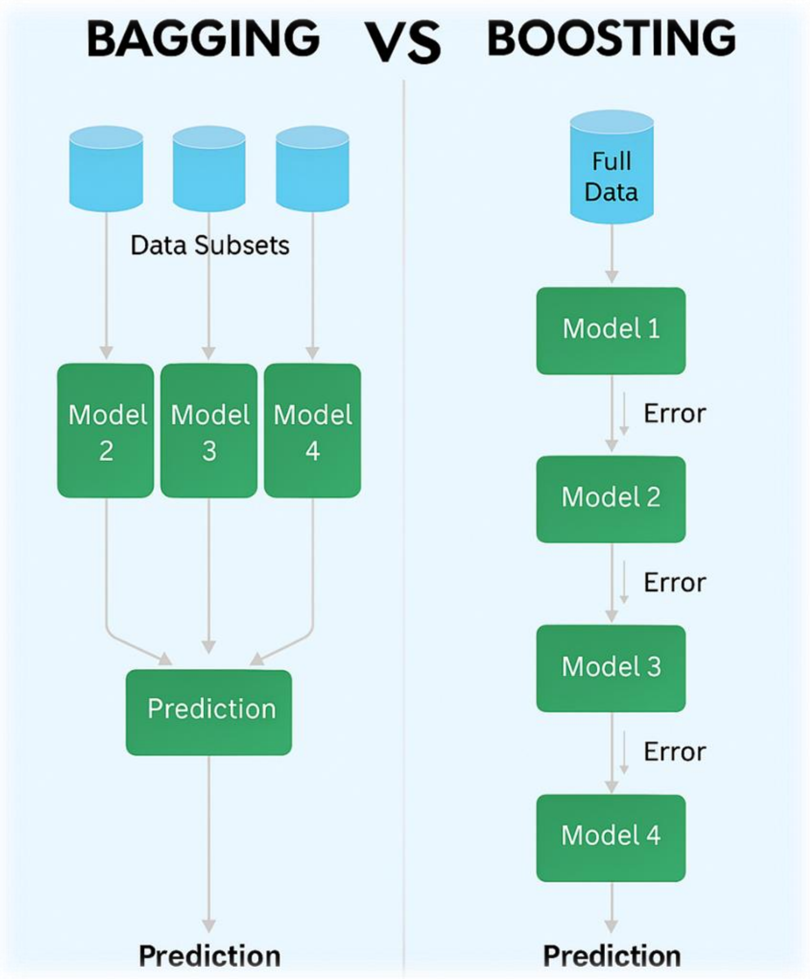
M1
0

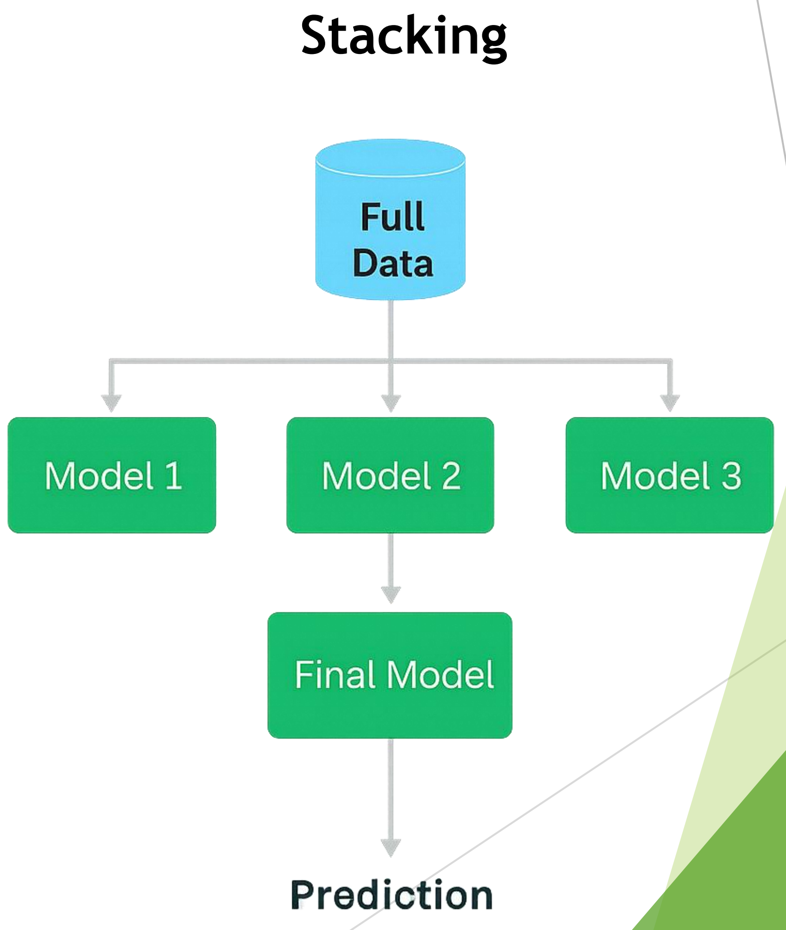
M2
1

M3
0

0

Comparison of Ensemble Learning Methods(review)





Naive Bayes

- Formula

$$P(y \mid x_1) = \frac{P(x_1 \mid y) \cdot P(y)}{P(x_1)}$$

- We want to prove the equation
  - Conditional Probability
    - **The probability of target** y **given feature** x_1 **is the joint probability of both happening, divided by the probability of the feature**

$$P(y \mid x_1) = \frac{P(y \cap x_1)}{P(x_1)}$$

$$P(y \cap x_1) = P(x_1 \mid y) \cdot P(y)$$

  - If you substitute above equation

$$P(y \mid x_1) = \frac{P(x_1 \mid y) \cdot P(y)}{P(x_1)}$$

Naive Bayes

- General equation

$$P(y \mid x_1, x_2, \ldots, x_n) = \frac{P(y) \cdot P(x_1 \mid y) \cdot P(x_2 \mid y) \cdots P(x_n \mid y)}{P(x_1, x_2, \ldots, x_n)}$$

- You can download the script from here

Types of Naive Bayes models

Model Type	Feature Type	Assumed Distribution
Gaussian	Continuous numeric	Normal (Gaussian)
Multinomial	Discrete counts	Multinomial
Bernoulli	Binary (0/1)	Bernoulli

Covariance Reveals Feature Relationships

- Covariance shows how two features vary together
  - Positive
    - both increase together
  - Negative
    - one increases, the other decreases
  - Zero
    - no clear relationship
- You can use covariance between the target and features if the **target** is **continuous**.

Parameters vs hyperparameters

- **parameters are the coefficients**
  - that the model learns from the data during training
- **hyperparameters**
  - Process of choosing best model parameters call **hyperparameter tunning**
  - **settings you choose before training** a machine learning model.like
    - C, kernel and gamma for SVM
    - n_estimators for Random Forest

gridsearchcv

- helps you **find the best hyperparameters** for a machine learning model
- **Loop through models — GridSearchCV tunes each one.**

Assignment

- Load breast-cancer.csv
- Calculate covariance matrix
- Tell I having target variable is effective or not(any effect)
- Then use gridsearchcv and different models and find best hyperparameters for each